

$$\begin{aligned}
\hat{L}_i^2 &= \epsilon_{ijk} \epsilon_{ilm} \hat{x}_j \hat{p}_k \hat{x}_l \hat{p}_m \\
&= (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) \hat{x}_j \hat{p}_k \hat{x}_l \hat{p}_m \\
&= \hat{x}_j^2 \hat{p}_k^2 - \hat{x}_j \hat{p}_j \hat{x}_k \hat{p}_k,
\end{aligned} \tag{A.200}$$

which becomes $\hat{L}^2 = r^2 \hat{p}^2 - (\vec{r} \cdot \vec{\hat{p}})^2$. Therefore, we obtain a relationship between $\hat{p}^2$ and $\hat{L}^2$, which can be rewritten as:

$$\hat{p}^2 = \frac{\hat{L}^2 + (\vec{r} \cdot \vec{\hat{p}})^2}{r^2}. \tag{A.201}$$

From the above result, we can rewrite the Hamiltonian of a three-dimensional harmonic oscillator in terms of the angular momentum operator:

$$\hat{\mathcal{H}} = \frac{\hat{p}^2}{2m} + \frac{m}{2} \omega^2 r^2 \Rightarrow \hat{\mathcal{H}} = \frac{\hat{L}^2}{2mr^2} + \frac{(\vec{r} \cdot \vec{\hat{p}})^2}{2mr^2} + \frac{m}{2} \omega^2 r^2. \tag{A.202}$$

However, the problem requests to write the Hamiltonian in terms of the angular momentum operator (task accomplished above), and radial contributions only. Therefore, we need to consider the differential form of the linear momentum operator $\vec{\hat{p}} = -i\hbar \vec{\nabla}$. Considering:

$$\vec{\nabla} = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \tag{A.203}$$

we obtain:

$$\vec{r} \cdot \vec{\hat{p}} = -i\hbar r \frac{\partial}{\partial r} \Rightarrow (\vec{r} \cdot \vec{\hat{p}})^2 = -\hbar^2 r \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right). \tag{A.204}$$

From the above, we obtain the final Hamiltonian:

$$\hat{\mathcal{H}} = \frac{\hat{L}^2}{2mr^2} - \frac{\hbar^2}{2m} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{m}{2} \omega^2 r^2. \tag{A.205}$$

2. Eq. (A.205) above has three terms: one depends only on $\hat{L}^2$, but we know that $[\hat{L}^2, \hat{L}_z] = 0$. The other two terms have only radial contributions, which therefore also commute with $\hat{L}_z$. Thus, we can state that $[\hat{\mathcal{H}}, \hat{L}_z] = 0$.

A.7 Chapter 7

Conceptual exercises

Solution 7.1. One reason to expand the Hilbert space is to enable mathematical operations involving matrices that represent operators of different dimensions. For example, consider the Hamiltonian $\hat{\mathcal{H}} = \hat{S}_z^2$ (where

$\hat{S}_z = \hat{S}_{1z} + \hat{S}_{2z}$), for $s_1 = 1/2$ and $s_2 = 1$. How to add a 2×2 matrix to another 3×3 ? Only if we expand the Hilbert space such that both matrices have the same dimension 6×6 .

Solution 7.2. The first Hund's rule defines in which orbitals to place electrons of a multi-electron atom. Due to the Coulombian repulsion, we cannot place two electrons in the same orbital when other orbitals are available. This rule maximizes the value of the effective atomic spin. The second Hund's rule, on the other hand, maximizes the value of orbital angular momentum. Let's consider a Bohr atom, for instance. If two electrons are orbiting this nucleus, Coulombian repulsion is minimized if their orbits are in the same direction. This classical view helps to understand why the effective angular momentum of the atom should be maximized.

Solution 7.3. The spin-orbit coupling, governed by the Hamiltonian $\zeta \vec{S} \cdot \vec{L}$, regulates Hund's third rule. For a less/more than half-filled shell, the spin-orbit coupling parameter is positive/negative. Therefore, for a less than half-filled shell, the energy is minimized when the vectors $\vec{S}$ and $\vec{L}$ are anti-parallel (which implies the ground state $j = |l - s|$); whereas for a more than half-filled shell, the energy is minimized when the vectors $\vec{S}$ and $\vec{L}$ are parallel (which implies the ground state $j = l + s$).

Practice exercises

Solution 7.6. 1. $[\hat{L}^2, \hat{L}_{1z}] = 2i\hbar(\hat{L}_{1x}\hat{L}_{2y} - \hat{L}_{1y}\hat{L}_{2x}) \neq 0$.

2. $[\hat{L}^2, \hat{L}_{2z}] = -[\hat{L}^2, \hat{L}_{1z}]$.

3. Due to the results of the above items, it is possible to say that the observables do not share the same basis vector, since $[\hat{L}^2, \hat{L}_{1z}] \neq 0$.

Solution 7.7. 1. $m = m_1 + m_2$.

2. The local basis for two angular momenta can be written as $\{|l_1, l_2, m_1, m_2\rangle\}$, while the coupled basis can be written $\{|l_1, l_2, l, m\rangle\}$. Thus, it is possible to write:

$$\langle m_1, m_2 | L_z | l, m \rangle = \langle m_1, m_2 | (L_{1z} + L_{2z}) | l, m \rangle, \quad (\text{A.206})$$

and therefore:

$$(m - m_1 - m_2)\hbar\langle m_1, m_2 | l, m \rangle = 0. \quad (\text{A.207})$$

Note that the internal product $\langle m_1, m_2 | l, m \rangle$ is the Clebsch–Gordan coefficients, that are different from zero for $m - m_1 - m_2 = 0$, i.e., for $m = m_1 + m_2$.

Solution 7.8. The CG coefficients are obtained from the successive application of raising or lowering operators:

$$L_{\pm} | l, m \rangle = \sqrt{l(l+1) - m(m \pm 1)} | l, m \pm 1 \rangle. \quad (\text{A.208})$$

We know that $+l \geq m \geq -l$ and therefore the argument of the square root above is always positive, which leads to only real Clebsch–Gordan coefficients.

Solution 7.9. The problem considers two spins only, and therefore, their sum s can take values from $|s_1 - s_2|$ to $s_1 + s_2$. Thus: $3 \geq s \geq 1$, that is, $s = 1, 2, 3$. From this information, we can formulate the coupled basis $\{|s, m\rangle\}$, where the values of m are associated with each multiplet s . As for the local basis, we know the values of s_1 and s_2 , and therefore, their respective values for m_1 and m_2 are also known. Hence, Table A.1 can be formulated.

TABLE A.1 Vectors of the local and coupled basis, for $s_1 = 1$ and $s_2 = 2$. Table related to Exercise 7.9.

Coupled basis $\{ s, m\rangle\}$	Local basis $\{ m_1, m_2\rangle\}$
$ 3, +3\rangle$	$ +1, +2\rangle$
$ 3, +2\rangle$	$ +1, +1\rangle$
$ 3, +1\rangle$	$ +1, 0\rangle$
$ 3, 0\rangle$	$ +1, -1\rangle$
$ 3, -1\rangle$	$ +1, -2\rangle$
$ 3, -2\rangle$	$ 0, +2\rangle$
$ 3, -3\rangle$	$ 0, +1\rangle$
$ 2, +2\rangle$	$ 0, 0\rangle$
$ 2, +1\rangle$	$ 0, -1\rangle$
$ 2, 0\rangle$	$ 0, -2\rangle$
$ 2, -1\rangle$	$ -1, +2\rangle$
$ 2, -2\rangle$	$ -1, +1\rangle$
$ 1, +1\rangle$	$ -1, 0\rangle$
$ 1, 0\rangle$	$ -1, -1\rangle$
$ 1, -1\rangle$	$ -1, -2\rangle$

Note: (i) the vectors of the local and coupled basis were arranged in a table side by side, and there is no direct correspondence between vectors in the same row, which are placed like this for presentation convenience. (ii) Both bases have the same dimension of the Hilbert space, as expected: $\sum_{s=1}^3 (2s + 1) = (2 \times 1 + 1) + (2 \times 2 + 1) + (2 \times 3 + 1) = 15$ (coupled basis) and $\prod_{i=1}^2 (2s_i + 1) = (2 \times 1 + 1)(2 \times 2 + 1) = 15$ (local basis). Finally, (iii) we have used a simplified notation: $|s_1, s_2, s, m\rangle = |s, m\rangle$ (for vectors of the coupled basis) and $|s_1, s_2, m_1, m_2\rangle = |m_1, m_2\rangle$ (for vectors of the local basis).

Solution 7.10. This problem is somewhat more complex than the previous one, and we need to make a diagram for guidance, similar to what was done in the text of this chapter (see Fig. 7.1). Table A.2 contains these diagrams and the respective vectors requested by the problem. Note we are using a simplified notation: $|s_1, s_2, s_{12}, s_3, s, m\rangle = |s_{12}, s, m\rangle$ (for the vectors of the sequential basis) and $|s_1, s_3, s_{13}, s_2, s, m\rangle = |s_{13}, s, m\rangle$ (for the vectors of the non-sequential basis).

Sequential basis: $s_1 + s_2 \geq s_{12} \geq |s_1 - s_2|$ and $s_{12} + s_3 \geq s \geq |s_{12} - s_3|$;

Non-sequential basis: $s_1 + s_3 \geq s_{13} \geq |s_1 - s_3|$ and $s_{13} + s_2 \geq s \geq |s_{13} - s_2|$.

TABLE A.2 Sequential and non-sequential basis for three spins: $s_1 = 1$, $s_2 = 2$ and $s_3 = 1$. The number in parentheses indicates the dimension of the Hilbert subspace of the respective multiplet. Table related to Exercise 7.10.

Sequential basis $\{ s_{12}, s, m\rangle\}$		Non-sequential basis $\{ s_{13}, s, m\rangle\}$	
s_{12}	s	s_{13}	s
3	4	2	4
	3		3
	2		2
2	3		1
	2		0
1	2	1	3
	1		2
	0		1
		0	2
$ 3, 4, m\rangle$ (9)		$ 2, 4, m\rangle$ (9)	
$ 3, 3, m\rangle$ (7)		$ 2, 3, m\rangle$ (7)	
$ 3, 2, m\rangle$ (5)		$ 2, 2, m\rangle$ (5)	
$ 2, 3, m\rangle$ (7)		$ 2, 1, m\rangle$ (3)	
$ 2, 2, m\rangle$ (5)		$ 2, 0, m\rangle$ (1)	
$ 2, 1, m\rangle$ (3)		$ 1, 3, m\rangle$ (7)	
$ 1, 2, m\rangle$ (5)		$ 1, 2, m\rangle$ (5)	
$ 1, 1, m\rangle$ (3)		$ 1, 1, m\rangle$ (3)	
$ 1, 0, m\rangle$ (1)		$ 0, 2, m\rangle$ (5)	
Hilbert space size: 45		Hilbert space size: 45	

Solution 7.11. See previous exercises for further details about this solution. Note we are using a simplified basis, which only includes the varying quantum numbers of the system. Table A.3 contains the solution to the problem.

Solution 7.12. Let's begin solving this problem by writing the matrices that represent the components of angular momentum for the case under

TABLE A.3 Sequential and non-sequential basis for four spins: $s_1 = 1$ and $s_2 = s_3 = s_4 = 1/2$. The number in parentheses indicates the dimension of the Hilbert subspace of the respective multiplet. Table related to Exercise 7.11.

Sequential basis $\{ s_{12}, s_{13}, s, m\rangle\}$			Non-sequential basis $\{ s_{12}, s_{34}, s, m\rangle\}$		
s_{12}	s_{13}	s	s_{12}	s_{34}	s
3/2	2	5/2 3/2	3/2	1	5/2 3/2 1/2
	1	3/2 1/2	1/2	0	3/2 3/2 1/2 1/2
1/2	1	3/2 1/2			
	0	1/2			
$ 3/2, 2, 5/2, m\rangle$ (6)			$ 3/2, 1, 5/2, m\rangle$ (6)		
$ 3/2, 2, 3/2, m\rangle$ (4)			$ 3/2, 1, 3/2, m\rangle$ (4)		
$ 3/2, 1, 3/2, m\rangle$ (4)			$ 3/2, 1, 1/2, m\rangle$ (2)		
$ 3/2, 1, 1/2, m\rangle$ (2)			$ 3/2, 0, 3/2, m\rangle$ (4)		
$ 1/2, 1, 3/2, m\rangle$ (4)			$ 1/2, 1, 3/2, m\rangle$ (4)		
$ 1/2, 1, 1/2, m\rangle$ (2)			$ 1/2, 1, 1/2, m\rangle$ (2)		
$ 1/2, 0, 1/2, m\rangle$ (2)			$ 1/2, 0, 1/2, m\rangle$ (2)		
Hilbert space size: 24			Hilbert space size: 24		

consideration $s_1 = s_2 = 1/2$:

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{2} i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{A.209})$$

Following the procedure of Subsection 7.4, we can write the angular momentum component for each spin in the correct dimension:

$$\hat{S}_{1x} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (\text{A.210})$$

and

$$\hat{S}_{2x} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (\text{A.211})$$

Similarly, $\hat{S}_{1y}$, $\hat{S}_{2y}$, $\hat{S}_{1z}$, and $\hat{S}_{2z}$ can be written. However, for practical reasons, these will not be presented here, only the relevant products:

$$\hat{S}_{1x}\hat{S}_{2x} = \left(\frac{\hbar}{2}\right)^2 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad (\text{A.212})$$

$$\hat{S}_{1y}\hat{S}_{2y} = \left(\frac{\hbar}{2}\right)^2 \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \quad (\text{A.213})$$

$$\hat{S}_{1z}\hat{S}_{2z} = \left(\frac{\hbar}{2}\right)^2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{A.214})$$

Having obtained the above results, it is easy to write down the operator under consideration:

$$\hat{H} = \hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{A.215})$$

Regarding the vector, we know that:

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{A.216})$$

and, consequently:

$$|+, -\rangle = |+\rangle \otimes |-\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}. \quad (\text{A.217})$$

Solution 7.13. The operator in the statement can be written as:

$$\hat{H} = \hat{S}_z^2 = (\hat{S}_{1z} + \hat{S}_{2z} + \hat{S}_{3z})^2 \quad (\text{A.218})$$

and we need to determine $\hat{S}_{1z}$, $\hat{S}_{2z}$ and $\hat{S}_{3z}$. Thus:

$$\hat{S}_{1z} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \mathbb{1}_{1z} & 0 \\ 0 & -\mathbb{1}_{1z} \end{pmatrix} \quad (\text{A.219})$$

where $\mathbb{1}_{1z}$ is a 4×4 identity matrix. Similarly, we can calculate the other two operators of interest:

$$\hat{S}_{2z} = \frac{\hbar}{2} \begin{pmatrix} \mathbb{1}_{2z} & 0 \\ 0 & \mathbb{1}_{2z} \end{pmatrix}, \quad \text{where} \quad \mathbb{1}_{2z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (\text{A.220})$$

$$\hat{S}_{3z} = \frac{\hbar}{2} \begin{pmatrix} \mathbb{1}_{3z} & 0 \\ 0 & \mathbb{1}_{3z} \end{pmatrix}, \quad \text{where} \quad \mathbb{1}_{3z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (\text{A.221})$$

Finally, to obtain the operator, we simply sum the three contributions and square the matrix, leading to:

$$\hat{S}_z^2 = \frac{\hbar^2}{4} \begin{pmatrix} 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 9 \end{pmatrix}. \quad (\text{A.222})$$

Solution 7.14. 1. First, we need to write down the vectors of the local and coupled bases, which are detailed in Table A.4. Note (i) for the coupled basis, l ranges from $1/2$ to $3/2$ and (ii) the values $m_1 + m_2$ are presented in parentheses for better guidance, because the Clebsch–Gordan coefficients are non-zero only when the condition $m = m_1 + m_2$ is fulfilled. Thus, it is possible to write equations analogous to (7.90)–(7.93):

$$|3/2, +3/2\rangle = | +1/2, +1\rangle \quad (\text{A.223})$$

$$|3/2, +1/2\rangle = \langle +1/2, 0 | 3/2, +1/2 \rangle | +1/2, 0 \rangle + \langle -1/2, +1 | 3/2, +1/2 \rangle | -1/2, +1 \rangle \quad (\text{A.224})$$

$$|3/2, -1/2\rangle = \langle +1/2, -1 | 3/2, -1/2 \rangle | +1/2, -1 \rangle + \langle -1/2, 0 | 3/2, -1/2 \rangle | -1/2, 0 \rangle \quad (\text{A.225})$$

$$|3/2, -3/2\rangle = | -1/2, -1 \rangle \quad (\text{A.226})$$

$$|1/2, +1/2\rangle = \langle +1/2, 0 | 1/2, +1/2 \rangle | +1/2, 0 \rangle + \langle -1/2, +1 | 1/2, +1/2 \rangle | -1/2, +1 \rangle \quad (\text{A.227})$$

TABLE A.4 Vectors of the local and coupled basis for $l_1 = 1/2$ and $l_2 = 1$. Table related to Exercise 7.14.

Coupled basis $\{ l, m\rangle\}$	Local basis $\{ m_1, m_2\rangle\} (m_1 + m_2)$
$ 3/2, +3/2\rangle$	$ +1/2, +1\rangle (+3/2)$
$ 3/2, +1/2\rangle$	$ +1/2, 0\rangle (+1/2)$
$ 3/2, -1/2\rangle$	$ +1/2, -1\rangle (-1/2)$
$ 3/2, -3/2\rangle$	$ -1/2, +1\rangle (+1/2)$
$ 1/2, +1/2\rangle$	$ -1/2, 0\rangle (-1/2)$
$ 1/2, -1/2\rangle$	$ -1/2, -1\rangle (-3/2)$

$$|1/2, -1/2\rangle = \langle +1/2, -1|1/2, -1/2\rangle |+1/2, -1\rangle + \langle -1/2, 0|1/2, -1/2\rangle |-1/2, 0\rangle. \quad (\text{A.228})$$

2. From the above equations, we can easily construct the matrix $\hat{U}$, analogous to that in Eq. (7.95):

$$\hat{U} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \langle +1/2, 0|3/2, +1/2\rangle & 0 & \langle -1/2, +1|3/2, +1/2\rangle & 0 & 0 \\ 0 & 0 & \langle +1/2, -1|3/2, -1/2\rangle & 0 & \langle -1/2, 0|3/2, -1/2\rangle & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \langle +1/2, 0|1/2, +1/2\rangle & 0 & \langle -1/2, +1|1/2, +1/2\rangle & 0 & 0 \\ 0 & 0 & \langle +1/2, -1|1/2, -1/2\rangle & 0 & \langle -1/2, 0|1/2, -1/2\rangle & 0 \end{pmatrix}. \quad (\text{A.229})$$

Solution 7.15. 1. The vectors for the local and coupled bases are detailed in Table A.5, where $2 \geq l \geq 0$.

TABLE A.5 Vectors of the local and coupled basis, for $l_1 = l_2 = 1$. Table related to the Exercise 7.15.

Coupled basis $\{ l, m\rangle\}$	Local basis $\{ m_1, m_2\rangle\} (m_1 + m_2)$
$ 2, +2\rangle$	$ +1, +1\rangle (+2)$
$ 2, +1\rangle$	$ +1, 0\rangle (+1)$
$ 2, 0\rangle$	$ +1, -1\rangle (0)$
$ 2, -1\rangle$	$ 0, +1\rangle (+1)$
$ 2, -2\rangle$	$ 0, 0\rangle (0)$
$ 1, +1\rangle$	$ 0, -1\rangle (-1)$
$ 1, 0\rangle$	$ -1, +1\rangle (0)$
$ 1, -1\rangle$	$ -1, 0\rangle (-1)$
$ 0, 0\rangle$	$ -1, -1\rangle (-2)$

2. In this task, we must relate the local basis to the coupled basis. For this, we consider Eq. (7.63):

$$|l_{\max}, m_{\max}\rangle = |m_{1\max}, m_{2\max}\rangle, \quad (\text{A.230})$$

which, in our case, is:

$$|2, +2\rangle = |+1, +1\rangle, \quad (\text{A.231})$$

this being the first relationship between the vectors of the two bases. Now, we must apply $L_- = L_{1-} + L_{2-}$ to both sides of the above equation: L_- by the left and $L_{1-} + L_{2-}$ by the right. Thus:

$$|2, +1\rangle = \frac{1}{\sqrt{2}} [|0, +1\rangle + |+1, 0\rangle]. \quad (\text{A.232})$$

Applying again the lowering operator on both sides of the above equation we find:

$$|2, 0\rangle = \frac{1}{\sqrt{6}} [| -1, +1\rangle + 2|0, 0\rangle + |+1, -1\rangle]. \quad (\text{A.233})$$

Continuing, as we need successive applications of $L_- = L_{1-} + L_{2-}$, we obtain:

$$|2, -1\rangle = \frac{1}{\sqrt{2}} [| -1, 0\rangle + |0, -1\rangle]. \quad (\text{A.234})$$

Finally, applying the lowering operator once more, the last equivalence reads as:

$$|2, -2\rangle = | -1, -1\rangle. \quad (\text{A.235})$$

With this procedure, we managed to map all the vectors of the coupled basis to the local basis, but only for the multiplet $l = 2$; however, we still have two other multiplets: $l = 1$ and $l = 0$. Now, let's solve the problem for $l = 1$. We can propose:

$$|1, +1\rangle = a|0, +1\rangle + b|+1, 0\rangle \quad (\text{A.236})$$

because we know that the Clebsch–Gordan coefficients are non-zero only for $m = m_1 + m_2$. The issue now is to obtain the values of a and b . For this purpose, let's choose a vector from the coupled basis that we already know and that has the same value of m – this can be Eq. (A.232). Why choose a vector from the coupled basis that has the same m ? Because this state can be associated with the same vectors from the local basis, that is: $|0, +1\rangle$ and $|+1, 0\rangle$. When we take the inner product:

$$\langle 2, +1 | 1, +1 \rangle = 0, \quad (\text{A.237})$$

and considering the normalization condition $|a|^2 + |b|^2 = 1$, we conclude that:

$$|1, +1\rangle = \frac{1}{\sqrt{2}} [|0, +1\rangle - |+1, 0\rangle]. \quad (\text{A.238})$$

As done for the multiplet $l = 2$, we now apply the lowering operator $L_- = L_{1-} + L_{2-}$ to both sides of the equation above, and obtain:

$$|1, 0\rangle = \frac{1}{\sqrt{2}} [| -1, +1\rangle - |+1, -1\rangle]. \quad (\text{A.239})$$

Finally, one more application of L_- yields:

$$|1, -1\rangle = \frac{1}{\sqrt{2}} [| -1, 0\rangle - |0, -1\rangle]. \quad (\text{A.240})$$

Now, we are left with only the singlet ($l = 0$). As was done for the case of the triplet ($l = 1$), we can now assume that:

$$|0, 0\rangle = a|+1, -1\rangle + b|0, 0\rangle + c|-1, +1\rangle, \quad (\text{A.241})$$

because these are the vectors of the local basis that have non-zero Clebsch–Gordan coefficients, due to the condition $m = m_1 + m_2$. We have three unknowns and therefore need three equations to fully determine them: a , b and c . The first equation is the normalization condition $|a|^2 + |b|^2 + |c|^2 = 1$. The second is the inner product $\langle 1, 0|0, 0\rangle = 0$ (see Eq. (A.239)). The third is the inner product $\langle 2, 0|0, 0\rangle = 0$ (see Eq. (A.233)). After simple calculations, we reach the conclusion that:

$$|0, 0\rangle = \frac{1}{\sqrt{3}} [|+1, -1\rangle - |0, 0\rangle + |-1, +1\rangle]. \quad (\text{A.242})$$

3. From the results of item (2) above, it is easy to construct the Clebsch–Gordan matrix $\hat{U}$, which relates the coupled basis to the local basis through the relation $|l, m\rangle = \hat{U}|m_1, m_2\rangle$:

$$\hat{U} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/\sqrt{6} & 0 & 2/\sqrt{6} & 0 & 1/\sqrt{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1/\sqrt{2} & 0 & 0 & 0 & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1/\sqrt{3} & 0 & -1/\sqrt{3} & 0 & 1/\sqrt{3} & 0 & 0 \end{pmatrix}. \quad (\text{A.243})$$

Development exercises

Solution 7.16. The operator $\hat{H} = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2$ is not diagonal in the local basis $|s_1, s_2, m_1, m_2\rangle$. To change the basis, let's consider $\vec{\hat{S}} = \vec{\hat{S}}_1 + \vec{\hat{S}}_2$, which allows us to rewrite the operator under consideration as:

$$\hat{H} = \frac{1}{2} (\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2). \quad (\text{A.244})$$

Now, the operator $\hat{H}$ only depends on operators of the coupled basis and therefore:

$$\hat{H}|s_1, s_2, s, m\rangle = E_s|s_1, s_2, s, m\rangle, \quad (\text{A.245})$$

where

$$E_s = \frac{1}{2} [s(s+1) - s_1(s_1+1) - s_2(s_2+1)] \quad (\text{A.246})$$

and $\{|s_1, s_2, s, m\rangle\}$ are vectors of the coupled basis. To proceed, we need to determine the possible values of s . These values range from $|s_1 - s_2|$ to $s_1 + s_2$. Thus, this system has one singlet $s = 0$ and one triplet $s = 1$, since $s_1 = s_2 = 1/2$. Finally, from Eq. (A.246) and from the possible values of s , we find:

$$E_0 = -\frac{3}{4} \quad \text{and} \quad E_1 = \frac{1}{4}. \quad (\text{A.247})$$

Solution 7.17. This exercise is similar to the previous one. For the coupled basis:

$$\hat{L}_z^2|l, m\rangle = m^2\hbar^2|l, m\rangle \quad (\text{A.248})$$

where $l = 0, 1, 2$. On the other hand, for the local basis:

$$\begin{aligned} \hat{L}_z^2|m_1, m_2\rangle &= (\hat{L}_{1z}^2 + \hat{L}_{2z}^2 + 2\hat{L}_{1z}\hat{L}_{2z})|m_1, m_2\rangle \\ &= (m_1^2 + m_2^2 + 2m_1m_2)\hbar^2|m_1, m_2\rangle. \end{aligned} \quad (\text{A.249})$$

The eigenvalues for $\hat{H}$, obtained from the equations above, are in Table A.6.

TABLE A.6 Local and coupled bases considering $l_1 = l_2 = 1$, as well as the corresponding eigenvalues of the operator $\hat{H}$ and states degeneracy.

Coupled basis $\{ l, m\rangle\}$	eigenvalue ($\hbar^2$)	deg.	Local basis $\{ m_1, m_2\rangle\}$	eigenvalue ($\hbar^2$)	deg.
$ 2, \pm 2\rangle$	4	2	$ \pm 1, \pm 1\rangle$	4	2
$ 2, \pm 1\rangle, 1, \pm 1\rangle$	1	4	$ \pm 1, 0\rangle, 0, \pm 1\rangle$	1	4
$ 2, 0\rangle, 1, 0\rangle, 0, 0\rangle$	0	3	$ \pm 1, \mp 1\rangle, 0, 0\rangle$	0	3

Advanced exercises

Solution 7.18. First, let's consider:

$$\vec{S}_{12} = \vec{S}_1 + \vec{S}_2 \quad \Rightarrow \quad \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} (\hat{S}_{12}^2 - \hat{S}_1^2 - \hat{S}_2^2). \quad (\text{A.250})$$

Next, we shall also consider:

$$\vec{\hat{S}} = \vec{\hat{S}}_{12} + \vec{\hat{S}}_3 \Rightarrow \vec{\hat{S}}_{12} \cdot \vec{\hat{S}}_3 = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_3 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_3 = \frac{1}{2} (\hat{S}^2 - \hat{S}_{12}^2 - \hat{S}_3^2). \quad (\text{A.251})$$

Thus, the Hamiltonian of the problem can be written as:

$$\hat{\mathcal{H}} = -\frac{J_1}{2} (\hat{S}_{12}^2 - \hat{S}_1^2 - \hat{S}_2^2) - \frac{J_2}{2} (\hat{S}^2 - \hat{S}_{12}^2 - \hat{S}_3^2), \quad (\text{A.252})$$

thus being diagonal in the sequential coupled basis $|s_1, s_2, s_{12}, s_3, s, m\rangle$:

$$\hat{\mathcal{H}}|s_1, s_2, s_{12}, s_3, s, m\rangle = E(s_{12}, s)|s_1, s_2, s_{12}, s_3, s, m\rangle. \quad (\text{A.253})$$

It is easy to see that:

$$E(s_{12}, s) = -\frac{1}{2}(J_1 - J_2)s_{12}(s_{12} + 1) - \frac{J_2}{2}s(s + 1), \quad (\text{A.254})$$

where we subtracted the constant energy terms associated with the constants s_1 , s_2 and s_3 . We now need to determine the possible values of s_{12} and s , being therefore: $s_1 + s_2 \geq s_{12} \geq |s_1 - s_2|$ and $s_{12} + s_3 \geq s \geq |s_{12} - s_3|$. Thus, we can construct the vectors, as described in Table A.7.

TABLE A.7 Sequential coupled basis for $s_1 = s_2 = 1/2$ and $s_3 = 1$. Table related to the Exercise 7.18.

Sequential coupled basis $\{ s_{12}, s, m\rangle\}$
$ 1, 2, m\rangle$
$ 1, 1, m\rangle$
$ 1, 0, m\rangle$
$ 0, 1, m\rangle$

From the eigenvectors presented in Table A.7, we can write the final energy spectrum:

$$E(1, 2) = -(J_1 + 2J_2) \quad (\text{quintuplet}) \quad (\text{A.255})$$

$$E(1, 1) = -J_1 \quad (\text{triplet}) \quad (\text{A.256})$$

$$E(1, 0) = -J_1 + J_2 \quad (\text{singlet}) \quad (\text{A.257})$$

$$E(0, 1) = -J_2 \quad (\text{triplet}) \quad (\text{A.258})$$

Solution 7.19. 1. The Hamiltonian for this problem is given by:

$$\hat{\mathcal{H}} = -J_2(\vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2 + \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_3 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_3) - J_1(\vec{\hat{S}}_1 \cdot \vec{\hat{S}}_4 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_4 + \vec{\hat{S}}_3 \cdot \vec{\hat{S}}_4). \quad (\text{A.259})$$

2. Let's consider the sequential coupled basis:

$$\vec{\hat{S}}_{12} = \vec{\hat{S}}_1 + \vec{\hat{S}}_2 \Rightarrow \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2 = \frac{1}{2} (\hat{S}_{12}^2 - \hat{S}_1^2 - \hat{S}_2^2) \quad (\text{A.260})$$

$$\vec{\hat{S}}_{13} = \vec{\hat{S}}_{12} + \vec{\hat{S}}_3 \Rightarrow \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_3 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_3 = \frac{1}{2} (\hat{S}_{13}^2 - \hat{S}_{12}^2 - \hat{S}_3^2) \quad (\text{A.261})$$

$$\vec{\hat{S}} = \vec{\hat{S}}_{13} + \vec{\hat{S}}_4 \Rightarrow \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_4 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_4 + \vec{\hat{S}}_3 \cdot \vec{\hat{S}}_4 = \frac{1}{2} (\hat{S}^2 - \hat{S}_{13}^2 - \hat{S}_4^2). \quad (\text{A.262})$$

Thus, we realize that this Hamiltonian is diagonal in the sequential coupled basis: $|s_1, s_2, s_{12}, s_3, s_{13}, s_4, s, m\rangle$:

$$\hat{\mathcal{H}}|s_1, s_2, s_{12}, s_3, s_{13}, s_4, s, m\rangle = E(s_{12}, s_{13}, s)|s_1, s_2, s_{12}, s_3, s_{13}, s_4, s, m\rangle. \quad (\text{A.263})$$

3. This problem considers four spins $s = 1/2$ sequentially coupled. To avoid mistakes, it is advisable to construct a figure (Fig. A.4) to assist in building the vectors, which are detailed in Table A.8.

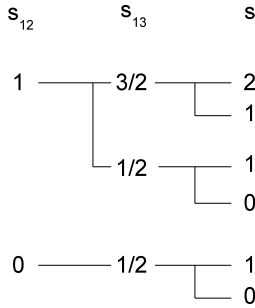


FIGURE A.4 Construction of the sequentially coupled basis for four spins $1/2$. Exercise 7.19.

TABLE A.8 Vectors of the sequential coupled basis, for four spins $1/2$. Exercise 7.19.

Sequential coupled basis $\{ s_{12}, s_{13}, s, m\rangle\}$
$ 1, 3/2, 2, m\rangle$
$ 1, 3/2, 1, m\rangle$
$ 1, 1/2, 1, m\rangle$
$ 1, 1/2, 0, m\rangle$
$ 0, 1/2, 1, m\rangle$
$ 0, 1/2, 0, m\rangle$

4. By using Eqs. (A.260)–(A.262) in the Hamiltonian of interest, we can rewrite it in the basis of item (2). Using Eq. (A.263), we can then obtain the energy of the system:

$$E(s_{12}, s_{13}, s, m) = E(s_{13}, s) = \frac{(J_1 - J_2)}{2} s_{13}(s_{13} + 1) - \frac{J_1}{2} s(s + 1). \quad (\text{A.264})$$

Note that above, we are subtracting from the system's energy all those constant terms dependent on s_1 , s_2 , s_3 and s_4 . From Eq. (A.264) and the vectors (see Table A.8), we can detail the energy spectrum as:

$$E(3/2, 2) = -\frac{3}{2}(J_1 + J_2) \quad (\text{quintuplet}) \quad (\text{A.265})$$

$$E(3/2, 1) = \frac{1}{2}(J_1 - 3J_2) \quad (\text{triplet}) \quad (\text{A.266})$$

$$E(1/2, 1) = -J_1 \quad (\text{triplet}) \quad (\text{A.267})$$

$$E(1/2, 0) = 0 \quad (\text{singlet}) \quad (\text{A.268})$$

$$E(1/2, 1) = -J_1 \quad (\text{triplet}) \quad (\text{A.269})$$

$$E(1/2, 0) = 0 \quad (\text{singlet}) \quad (\text{A.270})$$

For presentation purposes, the entire energy spectrum above has been further subtracted by a constant: $3(J_1 - J_2)/8$.

Solution 7.20. There are two unpaired electrons, and by the first Hund's rule: $s = 1$. By the second Hund's rule: $l = +3 + 2 = 5$. These two electrons in the f shell fill less than half of the shell, and therefore, by the third Hund's rule, the ground state is $j = |l - s| = 4$. The highest excited state is $j = l + s = 6$ and the first excited state (requested by the problem) is $j = 5$. In Russell–Saunders notation, the state of interest is therefore 3H_5 , where H corresponds to $l = 5$.

Solution 7.21. 1. According to Hund's first rule: $s = 1$. The d -subshell has five orbitals, and thus, according to Hund's second rule: $l = 3$. With these two quantum numbers, it is possible to determine that $4 \geq j \geq 2$; and, according to Hund's third rule, we determine that the multiplet of the ground state is the one with $j = 2$, due to the less than half-filled shell. Thus, we have the following multiplets:

$$\begin{aligned} &{}^3F_4 \text{ (last excited state)} \\ &{}^3F_3 \\ &{}^3F_2 \text{ (ground state)}. \end{aligned} \quad (\text{A.271})$$

2.

$$\hat{\mathcal{H}} = \zeta \vec{\hat{S}} \cdot \vec{\hat{L}} + \frac{\beta}{2}(\hat{J}_x^2 + \hat{J}_y^2). \quad (\text{A.272})$$

The spin-orbit term was considered because it is the term that governs the Hund's third rule and, consequently, the energy ordering of the multiplets identified by the total angular momentum $\vec{\hat{J}} = \vec{\hat{L}} + \vec{\hat{S}}$.

3. It is possible to write $\vec{S} \cdot \vec{L}$ in terms of $\hat{J}^2$, as well as the anisotropic Hamiltonian proposed, such that we can rewrite the Hamiltonian of the problem as:

$$\hat{\mathcal{H}} = \zeta \vec{S} \cdot \vec{L} + \frac{\beta}{2} (\hat{J}_x^2 + \hat{J}_y^2) \Rightarrow \hat{\mathcal{H}} = \frac{\zeta}{2} (\hat{J}^2 - \hat{S}^2 - \hat{L}^2) + \frac{\beta}{2} (\hat{J}^2 - \hat{J}_z^2). \quad (\text{A.273})$$

Thus, the Hamiltonian on the right-hand side is diagonal in the basis formed by the commuting operators $\{\hat{S}^2, \hat{L}^2, \hat{J}^2, \hat{J}_z\}$, whose vectors are represented as $\{|s, l, j, m\rangle\} \rightarrow \{|jm\rangle\}$.

4.

$$\hat{\mathcal{H}}|jm\rangle = E(j, m)|jm\rangle \quad (\text{A.274})$$

where

$$E(j, m) = \frac{\zeta}{2} j(j+1) + \frac{\beta}{2} [j(j+1) - m^2]. \quad (\text{A.275})$$

5.

$$\begin{aligned} E(2, m) &= 3\zeta + \beta \left(3 - \frac{m^2}{2}\right) \\ E(3, m) &= 6\zeta + \beta \left(6 - \frac{m^2}{2}\right) \\ E(4, m) &= 10\zeta + \beta \left(10 - \frac{m^2}{2}\right). \end{aligned} \quad (\text{A.276})$$

6. Substituting the possible values of m into the multiplets, we verify that the anisotropic term increases the energy of the system.

Solution 7.22. 1. To write the multiplets, we need Hund's rules. According to the first rule: $s = 7 \times \frac{1}{2} - \frac{1}{2} = 3$; and, by Hund's second rule: $l = +3 + 2 + 1 + 0 - 1 - 2 - 3 + 3 = 3$. We also know there are eight electrons in the shell (distributed in 14 possible states), thus the shell is more than half-full. Therefore, by Hund's third rule, the *ground* state is $j = l + s = 6$, while the last excited state is $j = |l - s| = 0$. The Russell–Saunders notation expresses the states as $^{2s+1}X_j$, and, in this case, we have:

$$\begin{aligned} &^7F_0 \text{ (last excited state)} \\ &^7F_1 \\ &^7F_2 \\ &^7F_3 \\ &^7F_4 \end{aligned}$$

$7F_5$

$${}^7F_6 \text{ (ground state)} \quad (\text{A.277})$$

where F represents $l = 3$.

2. Hund's third rule is governed by the spin-orbit coupling $\hat{\mathcal{H}} = \zeta \vec{S} \cdot \vec{L}$ where, in our case with a more than half-filled shell, the ζ parameter is smaller than zero. Thus:

$$\hat{\mathcal{H}} = -\zeta \vec{S} \cdot \vec{L} = -\frac{\zeta}{2} [\hat{J}^2 - \hat{S}^2 - \hat{L}^2], \quad (\text{A.278})$$

since $\vec{J} = \vec{L} + \vec{S}$. Note this procedure is a basis change, from the local one $|s, l, m_s, m_l\rangle$ to the coupled one $|s, l, j, m\rangle$. The above Hamiltonian is diagonal in the coupled basis, that is:

$$\hat{\mathcal{H}}|s, l, j, m\rangle = E(s, l, j)|s, l, j, m\rangle \quad (\text{A.279})$$

where

$$E(s, l, j) = E_j = -\frac{\zeta}{2} j(j+1). \quad (\text{A.280})$$

Note that above we are subtracting from the energy of the system constant terms that depend on l and s .

3. For each multiplet we have therefore:

$$\begin{aligned} E_0 &= 0 \text{ (last excited state)} \\ E_1 &= -\zeta \\ E_2 &= -3\zeta \\ E_3 &= -6\zeta \\ E_4 &= -10\zeta \\ E_5 &= -15\zeta \\ E_6 &= -21\zeta \text{ (ground state).} \end{aligned}$$

4. The energy difference between the ground state and the first excited state is 3.000 Kk_B , and therefore

$$|E_6 - E_5| = 6\zeta = 3.000 \text{ Kk}_B \Rightarrow \zeta = 500 \text{ Kk}_B. \quad (\text{A.281})$$

Thus, $E_6 = -21\zeta = -10.500 \text{ Kk}_B$.

Solution 7.23. 1. Let's consider the following sums:

$$\vec{\hat{S}}_{A'} = \vec{\hat{S}}_1 + \vec{\hat{S}}_3 \Rightarrow \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_3 = \frac{1}{2} (\hat{S}_{A'}^2 - \hat{S}_1^2 - \hat{S}_3^2) \quad (\text{A.282})$$

TABLE A.9 Non-sequential coupled basis for four spins 1/2. See Eqs. (A.282)–(A.285) for further details about how this basis was done. Exercise 7.23.

Non-sequential coupled basis $\{ s_{A'}, s_A, s_B, s, m\rangle\}$	
$ 1, 3/2, 1, 5/2, m\rangle$	
$ 1, 3/2, 1, 3/2, m\rangle$	
$ 1, 3/2, 1, 1/2, m\rangle$	
$ 1, 1/2, 1, 3/2, m\rangle$	
$ 1, 1/2, 1, 1/2, m\rangle$	
$ 0, 1/2, 1, 3/2, m\rangle$	
$ 0, 1/2, 1, 1/2, m\rangle$	
$ 1, 3/2, 0, 3/2, m\rangle$	
$ 1, 1/2, 0, 1/2, m\rangle$	
$ 0, 1/2, 0, 1/2, m\rangle$	

$$\vec{S}_A = \vec{S}_{A'} + \vec{S}_2 \Rightarrow \vec{S}_1 \cdot \vec{S}_2 + \vec{S}_3 \cdot \vec{S}_2 = \frac{1}{2} (\hat{S}_A^2 - \hat{S}_{A'}^2 - \hat{S}_2^2) \quad (\text{A.283})$$

$$\vec{S}_B = \vec{S}_4 + \vec{S}_5 \Rightarrow \vec{S}_4 \cdot \vec{S}_5 = \frac{1}{2} (\hat{S}_B^2 - \hat{S}_4^2 - \hat{S}_5^2) \quad (\text{A.284})$$

$$\vec{S} = \vec{S}_A + \vec{S}_B \Rightarrow \vec{S}_A \cdot \vec{S}_B = \frac{1}{2} (\hat{S}^2 - \hat{S}_A^2 - \hat{S}_B^2). \quad (\text{A.285})$$

Thus, we realize that the Hamiltonian of interest is diagonal in the coupled basis $|s_1, s_3, s_{A'}, s_2, s_A, s_4, s_5, s_B, s, m\rangle$ and thus, subtracting constants associated with s_1, s_2, s_3, s_4 , and s_5 , we can also obtain the energy of the system:

$$\begin{aligned} E(s_{A'}, s_A, s_B, s) = & -\frac{J_1}{2} [s_A(s_A + 1) - s_{A'}(s_{A'} + 1)] \\ & -\frac{J_2}{2} [s(s + 1) - s_A(s_A + 1) - s_B(s_B + 1)] \\ & -\frac{J_3}{2} [s_B(s_B + 1)]. \end{aligned} \quad (\text{A.286})$$

2. To write the eigenvalues, we need to know the vectors. For this purpose, what we have been doing is constructing an auxiliary figure, which is not presented here and we leave it for the reader to verify. Based on this figure, it is possible to construct the vectors, which are detailed in Table A.9. Thus, with the identified vectors and Eq. (A.286), it is possible to write the energy spectrum of the system:

$$E(1, 3/2, 1, 5/2) = -\frac{7}{8} J_1 - \frac{3}{2} J_2 - J_3 \quad (\text{sextuplet}) \quad (\text{A.287})$$

$$E(1, 3/2, 1, 3/2) = -\frac{7}{8} J_1 + J_2 - J_3 \quad (\text{quadruplet}) \quad (\text{A.288})$$

$$E(1, 3/2, 1, 1/2) = -\frac{7}{8}J_1 + \frac{5}{2}J_2 - J_3 \quad (\text{doublet}) \quad (\text{A.289})$$

$$E(1, 1/2, 1, 3/2) = \frac{5}{8}J_1 - \frac{1}{2}J_2 - J_3 \quad (\text{quadruplet}) \quad (\text{A.290})$$

$$E(1, 1/2, 1, 1/2) = \frac{5}{8}J_1 + J_2 - J_3 \quad (\text{doublet}) \quad (\text{A.291})$$

$$E(0, 1/2, 1, 3/2) = -\frac{3}{8}J_1 - \frac{1}{2}J_2 - J_3 \quad (\text{quadruplet}) \quad (\text{A.292})$$

$$E(0, 1/2, 1, 1/2) = -\frac{3}{8}J_1 + J_2 - J_3 \quad (\text{doublet}) \quad (\text{A.293})$$

$$E(1, 3/2, 0, 3/2) = -\frac{7}{8}J_1 \quad (\text{quadruplet}) \quad (\text{A.294})$$

$$E(1, 1/2, 0, 1/2) = \frac{5}{8}J_1 \quad (\text{doublet}) \quad (\text{A.295})$$

$$E(0, 1/2, 0, 1/2) = -\frac{3}{8}J_1 \quad (\text{doublet}). \quad (\text{A.296})$$

A.8 Chapter 8

Conceptual exercises

Solution 8.1. Let us consider a total Hamiltonian that is the sum of an unperturbed term (diagonal) and a perturbative one (non-diagonal). By diagonalizing this total Hamiltonian using the secular equation and then obtaining its respective vectors, we will observe that these vectors can be written in terms of various vectors of the unperturbed term. In other words, by adding a non-diagonal perturbation, these elements promote a mixing of the unperturbed states. The same happens in perturbation theory, where the first order correction $|n^{(1)}\rangle$ to the state $|n\rangle$ begins to exhibit this mixing of states. In short, the fact that the first order correction $|n^{(1)}\rangle$ to the state $|n\rangle$ depends on other states $|m^{(0)}\rangle$ means that the states mix after the perturbation. In addition, note that, according to Eq. (8.34), the closer these states are in (unperturbed) energy, the stronger the mixing will be, meaning that the dependence of $|n^{(1)}\rangle$ on $|m^{(0)}\rangle$ will be greater.

Text-based exercises

Solution 8.3.

$$\lambda^k : \quad \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(k)}\rangle = -\hat{\mathcal{V}} |n^{(k-1)}\rangle + \sum_{p=1}^k E_n^{(p)} |n^{(k-p)}\rangle, \quad k \geq 1. \quad (\text{A.297})$$

Note that $k = 0$ represents the unperturbed case and is not included in the recurrence above.